

# A robust and efficient citrus counting approach for large-scale unstructured orchards

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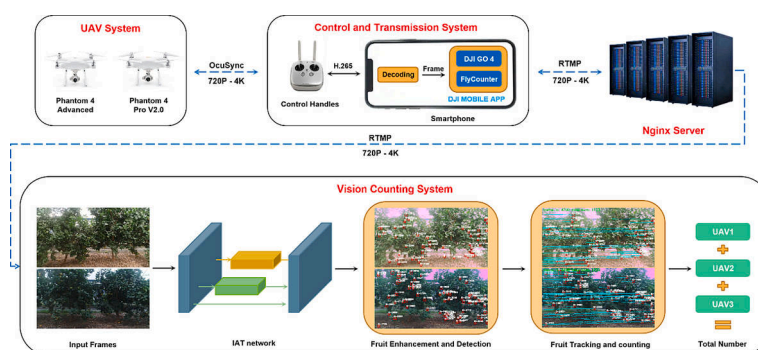
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## HIGHLIGHTS

- An End-to-end and efficient counting system for orchards using UAVs.
- A new UAV platform and flight application.
- A novel detection model is proposed for UAV scenarios.
- Introducing IAT network to improve the system's performance under low illumination.
- Improve work efficiency by using multiple UAVs to operate simultaneously.

## GRAPHICAL ABSTRACT



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## ABSTRACT

**CONTEXT:** Accurately detecting and counting fruits is essential for orchard yield estimation and smart management. However, two drawbacks are inherent in the current citrus counting algorithms: First, the robustness needs to be improved, particularly illumination changes and dense occlusion. Secondly, the actual operating efficiency of the system is low.

**OBJECTIVE:** To tackle the above issues, this paper proposed a robust and efficient fruit-counting pipeline based on Unmanned Aerial Vehicle (UAV).

**METHODS:** First, to obtain UAV video streaming data online, a live broadcast platform and a flight control Application called FlyCounter were developed. Secondly, the Illumination-Adaptive-Transformer network is used to enhance the low-illumination citrus image in real-time. Then, for the specific challenging scenario, a novel model named Fruit-YOLO is designed to accurately detect citrus in data streams. Finally, the DeepSORT is adopted to track and count fruits in video sequences.

**RESULTS AND CONCLUSIONS:** The results indicate that for detection performance, the P, R and mAP of the Fruit-YOLO of input enhanced image are 0.898, 0.854 and 0.929 respectively, which are 1.7%, 4.6% and 3.5% higher than the original image respectively. Regarding counting performance, for the daytime and evening scenes, the

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ID switch, MOTA and Error Mean of the pipeline are 63.7, 0.86 and 9.81% respectively. The MAPE of the offline and online operations of the pipeline are 4.36% and 12.66% respectively. The time and resources consumed by the system under parallel operations with different numbers of UAVs were analyzed.

**SIGNIFICANCE:** In this study, an online counting pipeline based on UAVs that can work in low-light scenarios is implemented for the first time. The system has good performance in both daytime and nighttime scenarios, enabling efficient counting in orchards and extending operating time.

## 1. Introduction

In recent years, fruit yield estimation, pest detection, and harvest robots have become a focus in the agricultural field (Xu et al., 2023; Ye et al., 2023; Wang et al., 2023). China plays an important role in the production and consumption of fruits in the world, with citrus planting area and output ranking first all year round. (Zheng et al., 2022; Meng et al., 2023). Accurately obtaining the number of citrus is a key prerequisite for intelligent orchard yield estimation. However, the task of counting fruits in orchards is a crucial yet demanding endeavor, primarily because it involves addressing various challenges such as severe occlusion, illumination changes, and fruits of varying sizes. The bottleneck problems of existing methods are: (1) The counting algorithm has poor robustness and low accuracy under dense occlusion and light changes. (2) In actual orchard operations, the overall system process is complex and inefficient. Therefore, developing a robust and efficient approach for fruit counting holds immense importance in advancing the realm of intelligent orchard yield estimation and ultimately enhancing the profitability of the industry.

With the advancement of the sensor, computer, and artificial intelligence technologies, some researchers have proposed various methods for estimating crop yields based on the quantity of fruits (Apolo-Apolo et al., 2020a, 2020b). Obtaining fruit quantity is the key to achieving intelligent orchard yield estimation, which involves the following three difficulties:

### (1) Work platform

In existing fruit-counting research, images are mostly collected through sensors held by people or mounted on vehicles (Ye et al., 2023; Shen et al., 2023; Vasconez et al., 2020). This means that this methods of operation remain less efficient and may cause soil damage problems. In comparison, Unmanned Aerial Vehicles (UAVs) with low cost, rich perspective information and high flexibility can better meet the operation requirements. Some scholars have effectively implemented the statistics of crops and animals by using UAVs (Zhang et al., 2023; Arakawa et al., 2023; Apolo-Apolo et al., 2020a, 2020b; Méndez et al., 2019; Rivas et al., 2018). Utilizing an aerial perspective for fruit counting represents a highly promising approach. However, due to the poor robustness of the algorithm in scenarios such as illumination changes, occlusion overlap, and small-sized fruits, it is difficult to apply to large-scale orchards.

Furthermore, the majority of previous studies obtain fruit counting results through offline processing of pre-recorded UAV videos. Such methods usually have the disadvantages of complex operational processes and lack of real-time execution capability. More importantly, previous research primarily concentrates on the development of yield estimation algorithms, with data collection being separated and offline, thus failing to achieve an efficient and end-to-end system. These methods cannot be adjusted in time to obtain the best estimation accuracy, which may ultimately result in the inability to estimate yields on a large scale in orchards within the specified time (Zheng et al., 2022).

### (2) Fruit detection

At present, significant advancements have been made by many researchers in the visual detection of citrus (Tang et al., 2023). Chen et al.

(2021) suggested a detection algorithm for citrus in different growth stages based on the improvement of YOLOv4. Zheng et al. (2021) proposed a method called YOLO BP for detecting green citrus. The method had an average accuracy of 86.00% in 6 complex scenarios. Huang et al. (2022) proposed a method for detecting green citrus by incorporating the use of CBAM in YOLOv5 and utilizing  $\alpha$ -IoU as the loss function. Their method achieved an average accuracy of 91.3% and a detection time of 16.7 ms. Chen et al. (2022) introduced a network for citrus detection using improved YOLOv4, but the real-time performance was poor. In summary, the above studies demonstrate the effectiveness of detecting citrus based on CNNs. However, there are currently few deep-learning methods for detecting citrus in UAV images. The detection performance of existing methods for immature green citrus in complex orchard scenes still cannot meet the requirements.

Most of the existing citrus detection works are conducted in ideal conditions with high-quality images. However, a variety of harsh lighting conditions such as backlighting, low illumination, and exposure often exist in unstructured orchards. The reduction of image quality is one of the important factors that significantly affect the performance of the detector (Zhou et al., 2022). In recent years, some researchers have conducted research on fruit detection under different lighting conditions in orchards. Zheng et al. (2021) introduced a network named YOLO BP for the detection of green citrus, achieving an accuracy of 92.50% in challenging backlighting conditions. Wang and He (2022) proposed an apple detection method called MS-ADS, which has a recall rate of 97.2% and 98.1% for apples with shadows on the surface and uneven lighting, respectively. Xu et al. (2023) implemented a visual recognition method for tomato picking robots using the YOLOv3-tiny network. In direct light, backlight and side light scenarios, the recall rates of this method are 91.52%, 83.37% and 96.11% respectively, with the worst performance under backlight conditions. Zhou et al. (2022) used the histogram equalization algorithm to process fruit images with uneven brightness and implemented the detection of *Camellia oleifera* based on YOLOv7. The above study discussed the impact of different lighting conditions on algorithm performance in daytime orchards. Compared with structured environmental conditions, the algorithm is less robust under complex lighting conditions in orchards, especially in scenes with insufficient light such as backlight and low illumination. In addition, to address this problem, existing methods often need to construct data sets of different lighting scenes to improve model performance, which is difficult to annotate and time-consuming.

Lighting changes in orchards during the day affect the accuracy of the algorithm. In contrast, the effects of environmental factors such as light changes and shadows can be effectively avoided in low-light environments (Xiong et al., 2020). In recent years, several researchers have conducted studies on fruit detection in low-illumination environments. Xiong et al. (2020) a network based on the improved YOLO v3 to detect citrus at night. Liang et al. (2020) used YOLO v3 and UNET to detect lychees and fruit stems in a nighttime environment. Fu et al. (2021) proposed a kiwi fruit detection algorithm named DY3TNet, which achieved the highest performance of 0.9005 under night-time LED lighting. Experimental results show that using image enhancement methods or simple lighting equipment can improve the performance of fruit detection in orchards. Zhang et al. (2023) implemented cherry detection during the day and night using the YOLOv4-Tiny network. The mAP in the nighttime test sets was 94.72%, which was higher than the 90.78% in the daytime test set. The above research shows that fruit detection in

low-illumination scenarios can achieve higher accuracy and extend the operation time, such as yield estimation UAV and fruit picking robots. Therefore, fruit detection and counting in low-illumination scenes in orchards is an important area worthy of research. However, existing related research often uses additional artificial light sources, which has shortcomings such as complicated and inconvenient operations (Liang et al., 2020). In addition, wild orchards are large and remote, and it is difficult for power equipment to meet the lighting conditions for all fruit trees. At present, there is a lack of research on low-light fruit detection in the agricultural field. To tackle the challenge of poor robustness of algorithms under backlight conditions during the day and low-light conditions in the evening, it is necessary to develop a set of image enhancement algorithms.

### (3) Fruit counting

Accurate fruit counting is a key prerequisite for smart orchard yield estimation. In recent years, a lot of scholars have conducted studies estimating individual trees or entire orchards. Gao et al. (2021) used YOLOv4-Tiny combined with the improved Hungarian algorithm to implement apple counting. However, this method can only process image sequences within 30 s. Zhang et al., (2022a) suggested a citrus counting pipeline based on OrangeYolo and OrangeSort, which achieved a mAP of 0.957. He et al. (2022) counted fruits based on a multi-feature matching algorithm and adopted a more reasonable MOTA evaluation index. However, the method only achieves an AP of 78.40, which has poor detection performance. Based on RGB-D, Rong et al., (2023) proposed an improved YOLOv5-4D combined with the ByteTrack to achieve tomato detection and counting.

Some researchers are exploring the integration of three-dimensional information from objects to improve the accuracy of fruit counting tasks. Nellithimaru and Kantor (2019) introduced the use of neural network models and SLAM to count grapes. Apolo-Apolo et al. (2020a, 2020b) used images to reconstruct and count fruits, and the method error was 6.59%. This method has good accuracy but requires more expensive software to process the image data pre-captured by the UAV. Hani et al. (2020) were the first to apply a combination of the Gaussian Mixed Model and the Structure-from-Motion (SfM) to accomplish fruit counting in orchards. Gené-Mola et al. (2020) and others proposed using SfM to reconstruct fruit trees and segment and count fruits based on Mask R-CNN. The above methods achieve good accuracy by combining three-dimensional information, but usually require expensive equipment and complex processing procedures. In addition, the specific steps of such methods require several hours or even more than ten hours to reconstruct the fruit, thus affecting the subsequent work process. Therefore, the above methods may not adequately fulfill the demands for efficient operation in the context of future smart orchards.

To tackle the aforementioned challenges, this study proposes a novel fruit-counting methodology approach utilizing UAV, facilitating accurate and efficient real-time counting of fruits in orchards. The main contributions of this paper can be summarized as follows:

- (1) To the best of our knowledge, this is the first work to realize an online and efficient counting pipeline for orchards in low-light scenarios based on UAVs. By introducing the IAT network, the problem of poor algorithm robustness under daytime backlight and evening low-light conditions is effectively addressed. Moreover, this method effectively addresses the issue of labeling low-light images, thereby providing valuable support for the implementation of intelligent operations involving various other fruits in low-light environments.
- (2) A UAV data transmission platform for orchards was established, and a flight control application called FlyCounter was developed to realize intelligent flight of UAVs and online acquisition of video streams.

- (3) A fruit detection network called Fruit-YOLO was proposed, which effectively tackled challenges related to the missed detection of small-sized fruits and overlapping occlusions.
- (4) An end-to-end counting system is implemented, which accelerates the acquisition of accurate results. This not only expedites the development, iterative improvement, and application of counting systems for developers but also furnishes farmers with timely and valuable information for their decision-making processes in the same year.

## 2. Materials

### 2.1. Image acquisition

This study uses a data set from an orchard in Baiyun District, Guangzhou City, Guangdong, China. The fruit variety is orange, and the skin color is yellow-green and green. The image acquisition experiment was conducted in April and June of 2023. Images were collected using PHANTOM 4 UAV (DJI Technology Co., Ltd., Shenzhen, China) with a resolution of  $5472 \times 3078$  pixels and a shooting distance of 1.5 m. The video collection method is shown in Fig. 1 in the *Supplementary Material*. The experiment collected citrus images at different angles, under different lighting conditions and at different times of the day. As shown in Fig. 2 in the *Supplementary Material*, there are complex situations such as overlapping occlusion, backlight, and low illumination in the evening. At the same time, the skin of some immature fruits is green.

### 2.2. Data pre-processing

In this study, a collection of 23 UAV videos was amassed, with each video encompassing 7 to 17 trees. Each of these trees bore a range of 80 to 100 fruits, and the video durations spanned from 4 to 8 min. The videos were randomly split into two sets: one with 13 videos for training and validation, and the other with 10 videos for testing. Since the training of the fruit recognition model requires feeding images, the frame extraction method (that is, extracting several frames from the video at certain intervals) was used to process the 13 training videos, and a total of 3954 training images were obtained. Finally, the dataset was divided into a training set and a validation set with a ratio of 7:3. The details of the division of citrus image data in each scene are shown in Table 1 in the *Supplementary Material*.

To reduce the cost of manpower and time, DarkLabel software is used (<https://github.com/darkpgmr/DarkLabel>) to annotate images. The annotation categories of ripe citrus and unripe citrus are “mature” and “immature” respectively. The XML file contains the category of each fruit and its normalized coordinates in the image.

## 3. Methodology

As shown in Fig. 1, the UAV-based fruit counting methodology consists of three components: hardware equipment system and flight control system, vision system. When carrying out the task, high-quality video streams are first collected by the UAV system and transmitted to the Android terminal device through the OcuSync protocol. Then, the flight control system pushes streaming media data to Nginx through the RTMP protocol. Finally, the visual counting system performs image enhancement, fruit detection and counting on the video stream.

### 3.1. Phase 1: UAV data transmission platform

The UAV control and transmission system is an important prerequisite for providing high-quality video streams online for subsequent vision systems. In previous experiments, we found that using a remote control handle has the following shortcomings: (1) The UAV's intelligent system prohibits flight when there are obstacles nearby, resulting in the UAV being unable to move in a narrow orchard. (2) The flight speed of

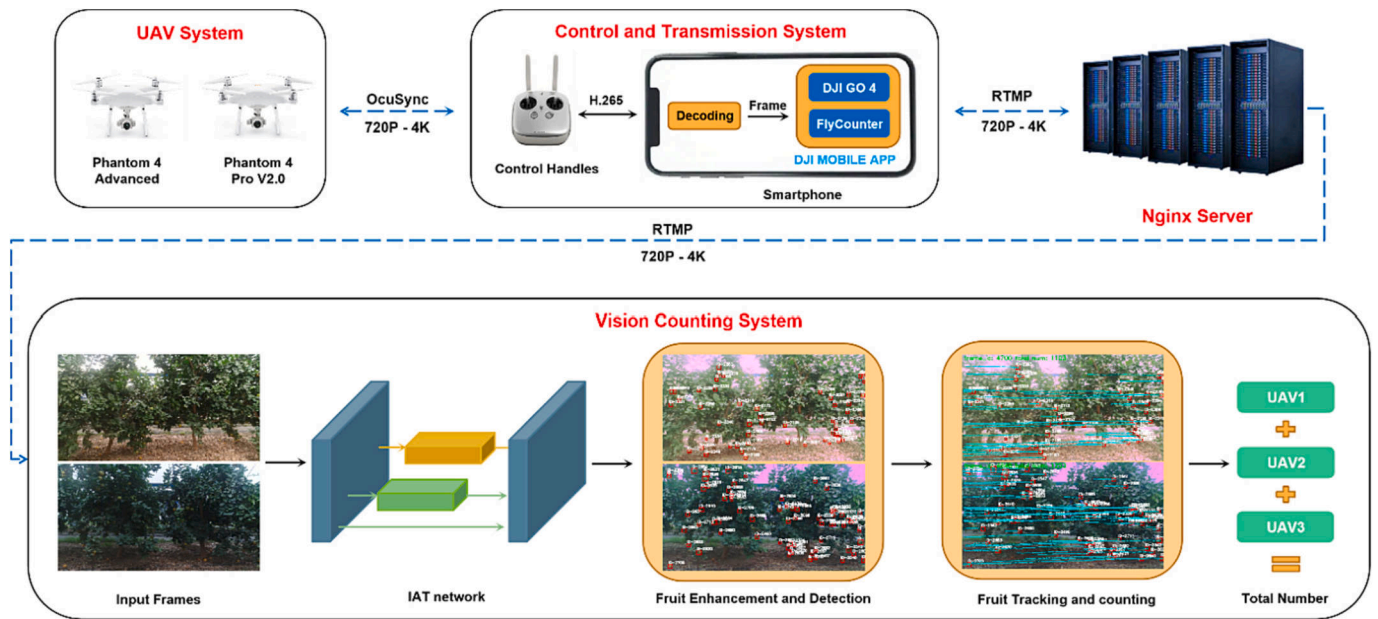


Fig. 1. The UAV-based citrus online counting pipeline.

UAVs under human control is difficult to maintain stable, which affects the performance of the algorithm.

To address the above problems, as shown in Fig. 2, an application called FlyCounter was developed to replace the previous manual control and realize the autonomous flight of UAVs. The “Enable/Close Virtual Stick” button provides the function of turning on or off the virtual stick, which is used for manual intervention to control the UAV in emergencies. The “Start/Stop Streaming” button provides the function of starting or stopping the UAV video stream. The “Take Off”, “Hover” and “Land” buttons provide the three basic control functions of takeoff, hovering and landing respectively. The “Accelerate/Decelerate” and “Time Increase/Decrease” buttons provide UAV flight speed and flight time control functions respectively. The obstacle avoidance system was turned off after the UAV took off to allow it to fly normally in the narrow environment of the orchard. Farmers can personalize the flight altitude, movement direction, flight speed and flight time on FlyCounter. FlyCounter controls the UAV to fly at a constant speed and photograph the orchard according to the set parameters, achieving intelligent flight of the UAV and real-time transmission of video streams.

With the continuous advancement of information and communications technologies, online live broadcasts have become popular. In this paper, a UAV video streaming online transmission platform for orchards is built based on the mainstream Nginx-RTMP-Module architecture. As illustrated in Fig. 3, the platform is composed of three components: push terminal, streaming media server, and pull terminal. These modules are sequentially interconnected via the RTMP protocol, with data transmission occurring between the streaming media end and the control end through LightBridge image transmission technology. In this study, the streaming terminals are UAV and FlyCounter applications. The lightweight Nginx server is adopted as the streaming media server. This server has the advantages of simple configuration and good scalability and is suitable for orchard field scenarios. The streaming end is a high-performance computing platform.

First, before the UAV takes off, the high-performance processor and mobile phone are connected to the same intranet through Wi-Fi. Then, set the focus mode of the UAV camera to automatic in FlyCounter to ensure the stability and clarity of the video stream. Then, the program re-encodes the video stream into a binary stream and transmits it to the port of the specified host according to the input streaming URL. Finally, the computing platform accesses the Nginx server through the RTMP

protocol to achieve online acquisition of video streams.

### 3.2. Phase 2: adaptive image enhancement

Although contemporary object detection models have excelled on various benchmark datasets, they encounter substantial difficulties when operating in low-light scenarios. In the complex environment of an orchard, there are many harsh lighting conditions, such as exposure, backlight, and insufficient light, which lead to a decrease in image quality and thus affect the performance of the detector. The Illumination-Adaptive-Transformer (IAT) network was proposed by Cui et al. (2022) to explore real-time dark light enhancement and visual tasks in a series of poor lighting scenarios. Compared with MAXIM (Tu et al., 2022) and Restormer (Zamir et al., 2021), the IAT network parameter amount is only 90 k+, which is an ultra-lightweight network that works well for counting orchard fruits in real-time. Therefore, the IAT model is used to enhance UAV images to address the problem of poor performance under backlight and low-illumination conditions.

Fig. 4 shows the IAT network structure diagram. The input citrus images will be processed through the Local branch and the Global branch to complete the low-light enhancement task. Among them, each local branch consists of three PEM (Pixel-wise Enhancement Module) modules. To ensure lightweight design and input at any resolution, a Transformer structure composed of depth-wise convolution is used. In the global branch, the Attention module is used to obtain global information to generate color matrices and gamma values. Enter the randomly initialized query into the module and work together with the key and value generated by the citrus image itself. Ten parameters are finally output, representing a  $3 \times 3$  color matrix and a 1-dimensional gamma value respectively. During model training, as the epoch increases, the network can adaptively adjust the color matrix and gamma value.

### 3.3. Phase 3: detection with deep learning

The accurate detection of citrus in the UAV video stream is a critical step in achieving orchard counting tasks. YOLOv5 (Ultralytics, 2021) has good real-time performance and accuracy, as well as fast fitting during training, making it suitable for the rapid development and deployment of systems for orchards. Nevertheless, during the orchard

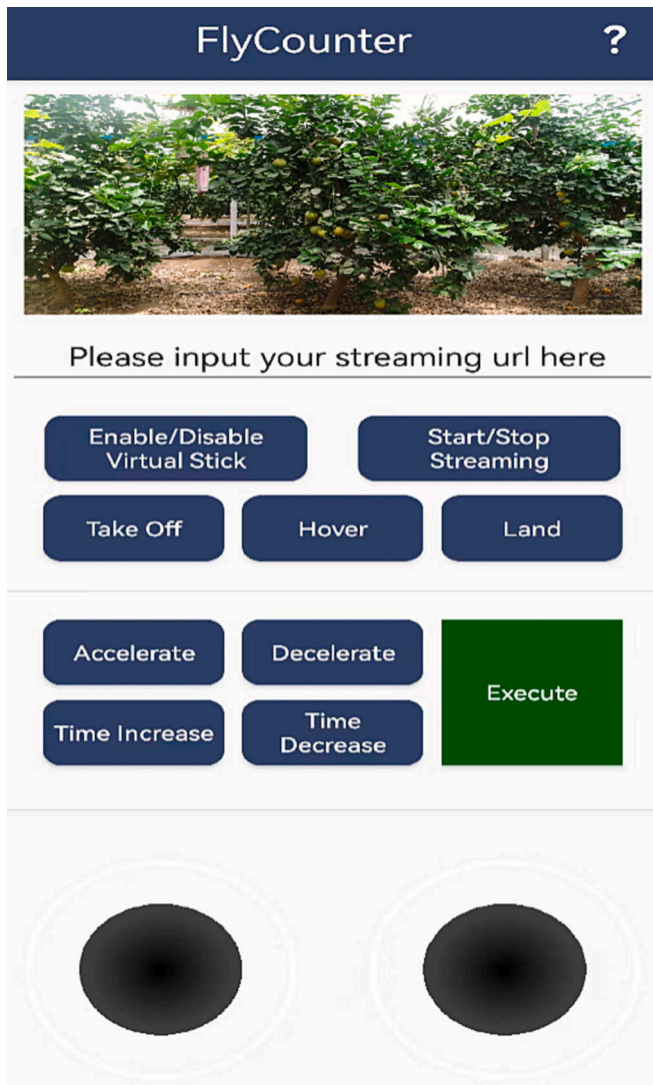


Fig. 2. FlyCounter APP.

field experiment, it became apparent that there is room for enhancement in the performance of the YOLOv5. In particular, high density, severe occlusion and overlap frequently occur in the orchard complex scene. At the same time, the features of immature fruits are similar to those of leaves. Therefore, to meet the accuracy requirements of the orchard counting task, the YOLOv5 needs further improvement.

Common designs in existing CNN architectures (i.e., across convolutional and/or pooling layers) are flawed, which leads to a

degradation of fine-grained information and less efficient feature representations to learn. To overcome the above limitations, we optimized YOLOv5 and proposed a citrus detection network called Fruit-YOLO (Fig. 5). The improvements are mainly in Backbone and Neck, using the SPD-Conv module to replace the original stride-2 convolution layer. The structure diagram of the SPD-Conv module is shown in Fig. 3 in the *Supplementary Material*. There are 7 such replacement instances, specifically in Backbone using 5 stride-2 convolutional layers to down-sample the feature map by a factor of 25, and in Neck using 2 stride-2 convolutional layers. Enhancing these aspects will contribute to an improved detection performance, particularly for small-sized objects within the model. At the same time, the shortcomings of occlusion, missing overlapping foreground objects and background object detection errors in the original YOLOv5 are effectively addressed.

### 3.4. Phase 4: fruit tracking and counting

After fruit pixel coordinate information is obtained by Fruit-YOLO, the DeepSORT algorithm is adopted to track and count fruits. Different from counting of a single frame image, the function of this part is to obtain the cross-frame correlation between the entire image sequence and ascertain the total count of fruits in the entire orchard. The Overview of Fruit tracking and counting is shown in Fig. 4 in the *Supplementary Material*.

#### 3.4.1. Kalman filter

In this study, the FPS of videos reached 24 frames per second, and fruits could be seen as uniformly moving. Therefore, fruits can be regarded as moving at a uniform speed, meeting the prerequisites for using the Kalman filter. The Kalman filter consists of two operations: trajectory prediction and update (Han et al., 2020).

For the trajectory forecasting operation, by utilizing the Fruit-YOLO's output from the k-1th frame image, the Kalman filter can forecast the precise location( $P'$ ) of the fruit in the k-th frame through Eq. (1) and Eq. (2).

$$\mathbf{x}' = \mathbf{F}\mathbf{x} + \mathbf{Q}_1 \quad (1)$$

$$\mathbf{P}' = \mathbf{F}\mathbf{P}\mathbf{F}^T + \mathbf{Q}_2 \quad (2)$$

For the update operation, the observations of the k-th image are used to optimize the prediction of the k-1-th image. Finally, update  $\mathbf{x}$  and  $\mathbf{P}$  through Eq. (6) and Eq. (7):

$$\mathbf{y} = \mathbf{z} - \mathbf{H}\mathbf{x}' \quad (3)$$

$$\mathbf{S} = \mathbf{H}\mathbf{P}'\mathbf{H}^T + \mathbf{R} \quad (4)$$

$$\mathbf{K} = \mathbf{P}'\mathbf{H}^T\mathbf{S}^{-1} \quad (5)$$

$$\mathbf{x} = \mathbf{x}' + \mathbf{K}\mathbf{y} \quad (6)$$

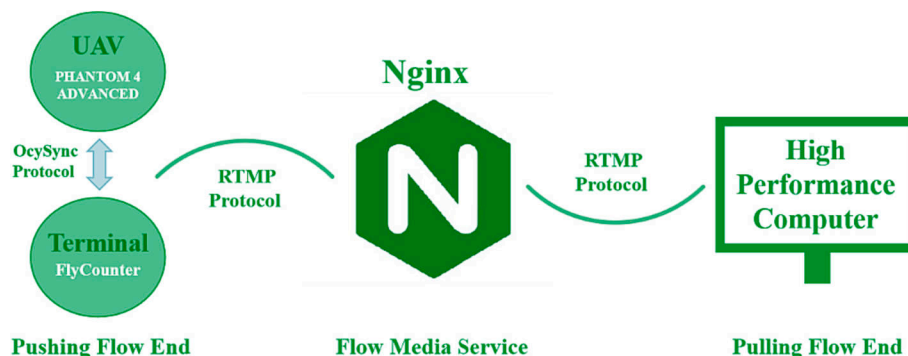


Fig. 3. The UAV live broadcast platform. It includes three modules: the push terminal, the streaming media server, and the pull terminal.

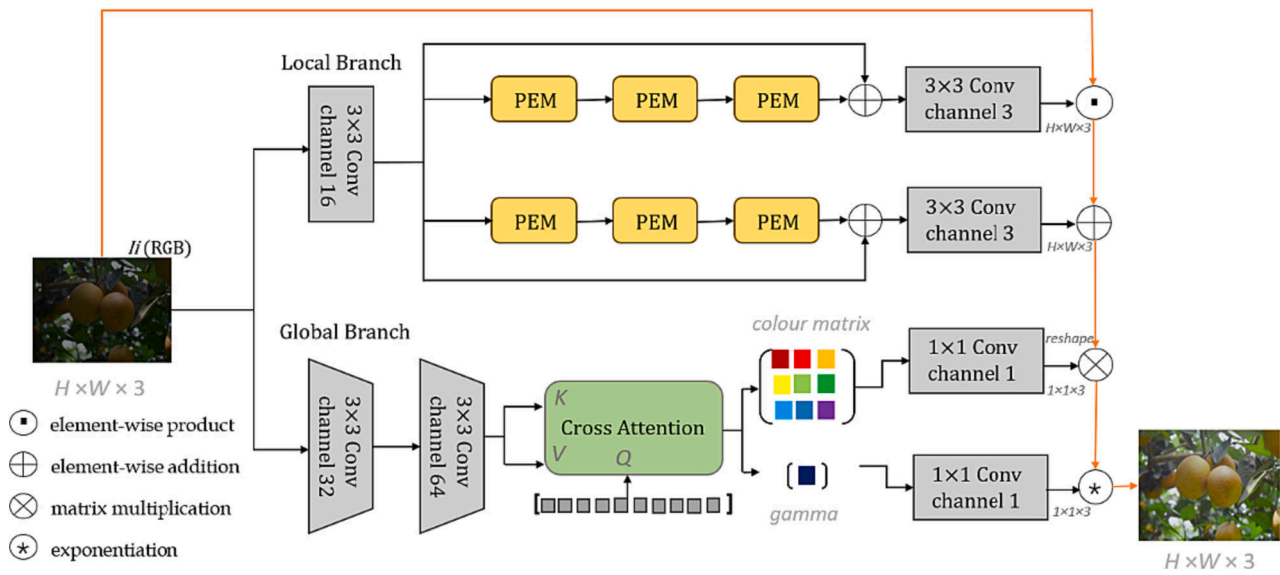


Fig. 4. Overview of the IAT network.

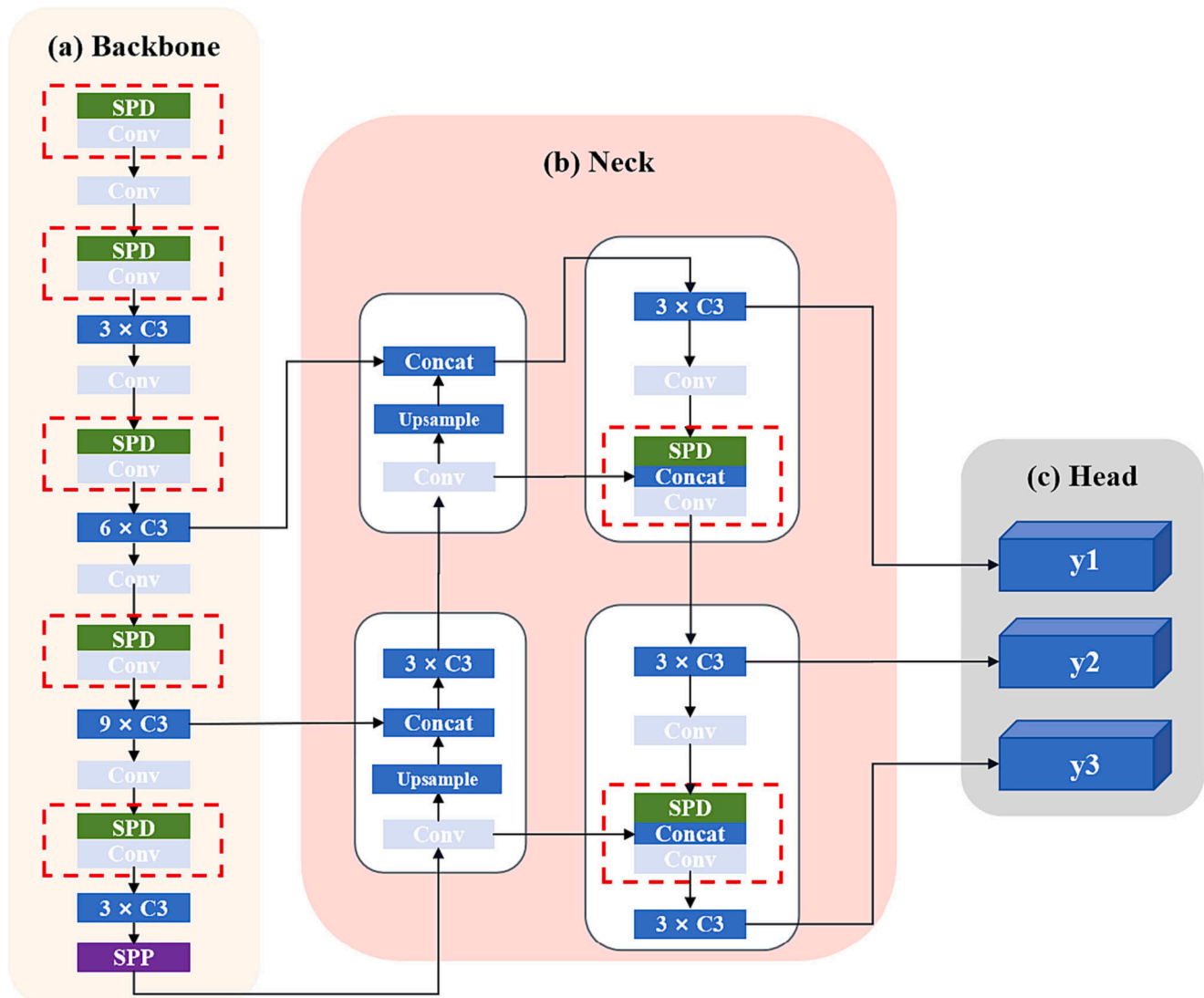


Fig. 5. Overview of Fruit-YOLO network. Red boxes are where the replacement happens. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$P = (I - KH)P \quad (7)$$

### 3.4.2. Hungarian algorithm

After obtaining the trajectory prediction information, the Hungarian algorithm is used to match the bounding box and track the trajectory in Cascade matching and IOU matching. Based on motion and appearance information, detection boxes and tracking predictions are matched by the Hungarian algorithm.

In the motion information, Mahalanobis distance is used to represent the matching score between bounding boxes and trajectories, as shown in Eq. (8). During the actual operation, if the user operates improperly or is affected by environmental factors, it will fail to track fruits only through Mahalanobis distance. To address the problems, the matching results can be further optimized by adding appearance information. In Eq. (10), compare the features of the  $i$ -th trajectory and the  $j$ -th bounding box, and calculate the minimum cosine distance. The two match when  $d^{(2)}(i, j)$  is lower than a certain value.

$$d^{(1)}(i, j) = (d_j - y_i) S_i^{-1} (d_j - y_i) \quad (8)$$

$$b_{ij}^{(1)} = 1[d^{(1)}(ij) \leq t^{(1)}] \quad (9)$$

$$d^{(2)}(i, j) = \min\{1 - r_j^T r_k^{(i)} | r_k^{(i)} \in R_i\} \quad (10)$$

To combine the two metrics, they are linearly weighted and used as a cascade matching measure. Finally, for the mismatched trajectories, detection boxes and newly appearing fruits, the Hungarian algorithm is used again for matching, and finally, fruits in the orchard are counted globally.

$$c_{ij} = \lambda d^{(1)}(i, j) + (1 - \lambda) d^{(2)}(i, j) \quad (11)$$

## 4. Experiments

### 4.1. Experimental environment

To fairly compare the performance of each algorithm, the training platform and hyperparameter settings are the same. The platform conditions are as follows: CPU is Intel Xeon Gold 6256, clocked at 3.60 GHz with 48 physical cores, 24 threads, and 1024 GB memory. The GPU is NVIDIA RTX A6000 with 48 GB display RAM. The system is Ubuntu 18.04, and the software includes CUDA 11.8.130, CUDNN 8.6.0, NVIDIA driver 535.104, Opencv 4.8.0 and training framework Pytorch 2.0.1.

### 4.2. Fruit detection experiment

#### 4.2.1. Model performance assessment

The error in fruit counting depends on the performance of the detector. In this section, an experiment is designed to verify the effectiveness of the proposed algorithm. The experimental subjects are YOLOv5 and Fruit-YOLO. Evaluation metrics include P, R, mAP and FPS.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

$$AP = \sum_{k=1}^N P(k) \Delta r(k) \quad (14)$$

where  $TP$  represents the correct sample,  $FP$  represents the incorrectly detected sample, and  $FN$  is the missed sample.  $AP$  is an indicator reflecting the accuracy of the detector.  $FPS$  is the time it takes for a single frame to be detected.

Figs. 6 and Table 1 shows the detection results and specific quantitative results of the two networks. In the UAV application scenario of

this study, Fruit-YOLO has successfully overcome the missed detection problems (overlapping occlusion, small size, and green citrus) and false detection problems (branches and leaves) of the networks of YOLOv5. Fruit-YOLO outperforms YOLOv5 in terms of P, R, and mAP, with an average mAP improvement of 1.20%. The detection speed of Fruit-YOLO is almost the same as that of YOLOv5. The results show that the Fruit-YOLO network has higher accuracy while ensuring real-time performance.

### 4.3. Image enhancement experiment

#### 4.3.1. Dataset construction and model training

Usually, the construction of this type of data set requires the use of cameras to collect fruit trees at fixed positions and heights under normal light and low illumination conditions, respectively. However, this requires a lot of equipment, labor and time costs. At the same time, it is difficult to manually annotate images collected under low-light conditions. In addition, it was found in actual tests that this method is not feasible due to the difficulty in maintaining the absolute position of the UAV in two different periods when flying. Therefore, this study constructs three gradients of fruit low-light data sets T1, T2 and T3. The gradient coefficients were set to 0.6, 0.35 and 0.1 respectively to simulate the actual light conditions in the orchard.

After IAT feeds the three data sets T1, T2 and T3 for training, three enhanced models MT1, MT2 and MT3 are obtained respectively. After iterating to 30 generations, the network is fitted. Judging from the parameter convergence, the network training results are ideal. The MT2 model performs best on the validation set, with PSNR and SSIM values approaching 40 and 1 respectively. The details of datasets and comparison results of enhanced models are shown in Fig. 5 and Table 3 in the *Supplementary Material*.

#### 4.3.2. Model performance assessment

In this section, an experiment was conducted to verify the feasibility of the IAT in the actual orchard. The test comparison objects include three low-light enhancement models, MT1, MT2 and MT3. The test set is 188 suitable light and low-illumination images taken by artificially fixing the height and position of the UAV in the orchard. PSNR (Peak Signal-to-Noise Ratio) and SSIM (Structural Similarity Index) are used as evaluation indicators.

$$PSNR = 10 \log_{10} \left( \frac{MAX^2}{MSE} \right) \quad (15)$$

$$SSIM(x, y) = \frac{(2\mu_x \mu_y + C_1) \cdot (2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1) \cdot (\sigma_x^2 + \sigma_y^2 + C_2)} \quad (16)$$

where  $MAX$  represents the maximum possible value of the signal.  $MSE$  (Mean Squared Error) represents the square of the average difference between the original signal and the distorted signal. The augmented image and ground truth are represented by  $x$  and  $x$  respectively. The average values of  $x$  and  $y$  are represented by  $\mu_x$  and  $\mu_y$  respectively. The variances of  $x$  and  $y$  are represented by  $\sigma_x$  and  $\sigma_y$  respectively.  $\sigma_{xy}$  represents the covariance of  $x$  and  $y$ .  $C_1$  and  $C_2$  are used to stabilize the calculation process.

The PSNR and SSIM of the MT2 are 26.571 and 0.795 respectively, which are 2.189 and 1.0% higher than the MT1, and 5.612 and 2.7% higher than the MT3 respectively. Experimental results show that the model trained on the T2 data set has the best performance. Therefore, in this study, the MT2 model was adopted for UAV image enhancement.

#### 4.3.3. Ablation experiment

The purposes and evaluation indicators of image enhancement algorithms and high-level visual algorithms are inconsistent. Image enhancement is to better improve human vision (PSNR, SSIM), while

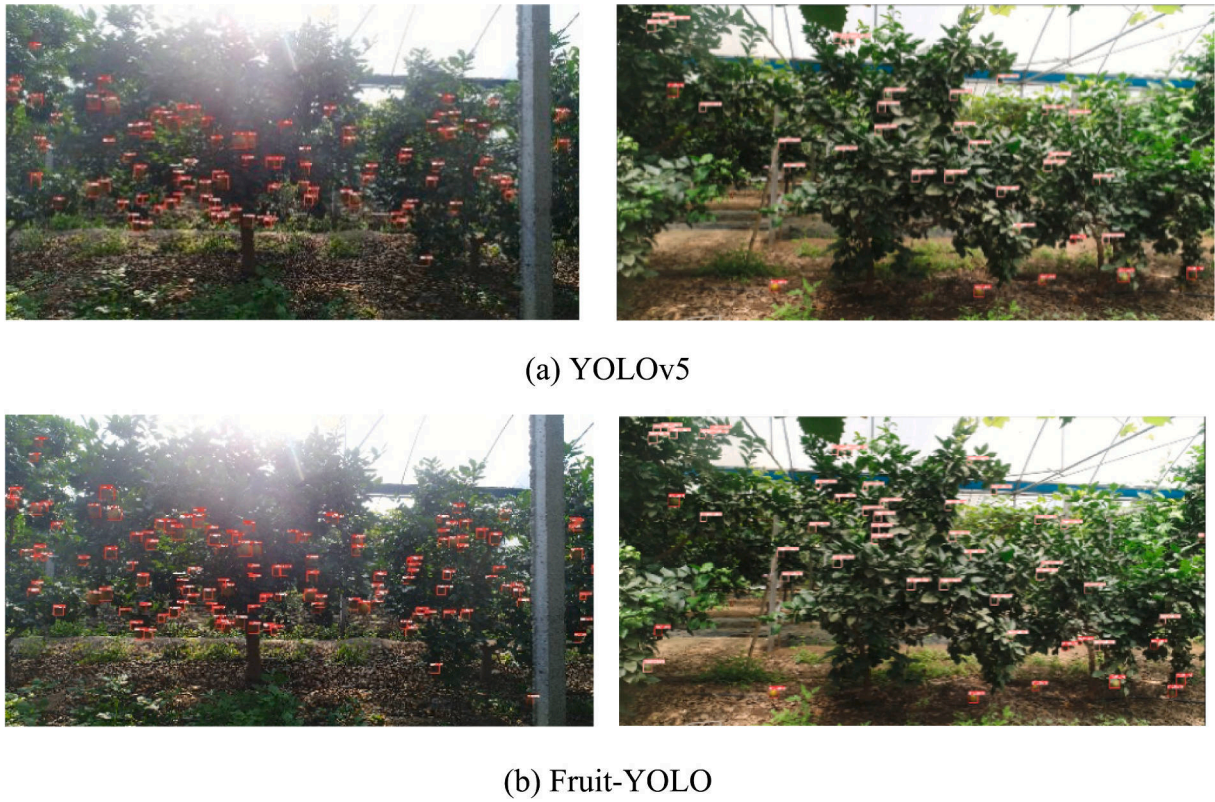


Fig. 6. Detection results of mature citrus and immature citrus.

**Table 1**  
Comparisons of three networks in a detection task.

| Class Label | P      |            | R      |            | mAP    |            | FPS    |            |
|-------------|--------|------------|--------|------------|--------|------------|--------|------------|
|             | YOLOv5 | Fruit-YOLO | YOLOv5 | Fruit-YOLO | YOLOv5 | Fruit-YOLO | YOLOv5 | Fruit-YOLO |
| Mature      | 0.874  | 0.879      | 0.878  | 0.889      | 0.910  | 0.921      |        |            |
| Immature    | 0.821  | 0.842      | 0.715  | 0.727      | 0.794  | 0.806      | 69     | 65         |
| Mean        | 0.848  | 0.861      | 0.797  | 0.808      | 0.852  | 0.864      |        |            |

object detection is a machine vision task (mAP). Therefore, an experiment is designed to explore whether the enhanced model can further improve the performance of Fruit-YOLO. Precision, recall, and mAP are adopted as evaluation metrics.

The P, R and mAP of the Fruit-YOLO inputting enhanced image are 0.878, 0.854 and 0.899 respectively, which are 1.7%, 4.6% and 3.5% higher than the original image respectively. After the UAV image is enhanced, citrus in the low-illumination area of the image can be detected by Fruit-YOLO. The result shows that the embedding of the IAT network improves the accuracy of the detector. The quantitative comparison results are shown in Table 4 in the *Supplementary Material*.

#### 4.3.4. Failure cases

The accuracy of the counting task depends heavily on the performance of the detector. However, in the unstructured environment of an orchard, the detector is easily affected by multiple factors, such as occlusion, light changes, and fine-grained feature changes (He et al., 2022). Although the problems of missed detection of small-sized citrus, overlapping occlusion and low-light fruits are well addressed by Fruit-YOLO and IAT, it also bring some problems.

As shown in Fig. 6 in the *Supplementary Material*, fruits in the red circle failed to be detected in the enhanced image. The reason is that under sufficient light conditions during the day, the high-brightness area at the edge of the image may be exposed due to the enhancement of the IAT network, resulting in missed detection of the Fruit-YOLO network.

In addition, leaves may be mistakenly detected as positive samples by the Fruit-YOLO. Therefore, the recall rate of the Fruit-YOLO network has increased by 4.6%, while the precision rate has only increased by 1.7%. The development of image area brightness evaluation and adaptive enhancement algorithms can be effective in addressing this problem.

#### 4.4. Fruit counting experiment

##### 4.4.1. Model performance assessment

In this section, an experiment was designed to compare the pipeline with YOLOv5 to verify the superiority of the proposed method. The same images of training, validation and testing are used in the experiments. L1 Loss, MOTA (Bernardin and Stiefelhagen, 2008), ID switch and Error Mean are used to evaluate counting performance.

$$\text{MOTA} = 1 - \frac{\sum_i (m_i + fp_i + mme_i)}{\sum_i gt_i} \quad (17)$$

$$\text{L1 Loss} = \text{Count} - \text{Ground Truth} \quad (18)$$

$$\text{Error Mean} = \frac{\text{L1 Loss}}{N} \quad (19)$$

where the ID switch represents the number of mismatches.  $GT_t$  is the number of fruits calculated manually. ID switch and MOTA reflect the local performance of the counter. MOTA can be seen as the ratio of er-

rors from  $FN_t$ ,  $FP_t$  and ID switch. Furthermore, we compare the results of the pipeline with the ground truth to evaluate the overall performance Count represents the result of the algorithm, Ground Truth is the manual calculation of total fruits, and N is the number of fruit trees.

Tables 2 and 3 show the specific quantitative results of the counting performance of the three methods in daytime and evening scenarios. For local tracking performance, the average ID switch and MOTA values of our method are 63.7 and 0.86 respectively, which are both lower than YOLO\_DSORT. For the overall counting performance, the Error Means of our method in the two scenarios are 8.58 and 11.04 respectively, which are lower than 12.03 and 13.33 of YOLO\_DSORT. Experimental results show that the proposed pipeline has better counting performance.

Different from previous single-frame counting tasks, the purpose of this study was to complete the counting of fruits in the entire orchard. Therefore, these metrics are calculated manually on a frame-by-frame basis for the entire video stream.

#### 4.4.2. Comparison of offline and online counting

Online fruit counting in orchards is realized through the established UAV live broadcast platform. However, network quality fluctuations in wild orchards will have an impact on the data transmission of the platform. To further analyze the impact of the platform on counting pipeline, an experiment based on offline and online counting method was conducted. The test subjects were three rows of fruit trees, including a total of 54 trees. Data distortion rate, ID Switch and MAPE are adopted as evaluations.

$$MAPE = \frac{1}{m} \sum_{j=1}^m \left| \frac{Count_j - GT_j}{GT_j} \right| \quad (20)$$

The comparison results are shown in Table 5 in the Supplementary Material. The method exhibits a delay ranging from 7 to 10 s, and the data distortion rate is 2.8%. In terms of counting performance, the method running offline demonstrates an average ID Switch of 26 and a MAPE of 4.36%, while the method running online increases to 65.53 and 12.66% respectively.

#### 4.4.3. Multi-UAV field experiment

Most previous methods often use indirect metrics to indicate efficiency, such as FLOPs or parameters. However, these metrics are sub-optimal and not suitable for deployment in real-world operational scenarios. As stated in a lot of works, existing evaluation metrics for designing neural networks often do not truly reflect direct measurements (Xia et al., 2022). In practice, we are more concerned with the FPS or latency of an algorithm besides performance.

Multiple UAVs can help improve the efficiency and promotion value of the counting pipeline. To further improve system efficiency, this paper explores the feasibility of parallel operations with different numbers of UAVs. The UAVs are implemented by multi-threaded concurrent execution. The experimental objects are the detection, enhancement and tracking of the counting pipeline. The content of the

**Table 3**

Comparison of overall performance.

| Method               | Daytime (77 Trees) |           | Evening (69 Trees) |           |
|----------------------|--------------------|-----------|--------------------|-----------|
|                      | YOLO_DSORT         | Ours      | YOLO_DSORT         | Ours      |
| Count / Ground Truth | 6991/6065          | 6726/6065 | 7877/6957          | 7719/6957 |
| L1 Loss              | 926                | 661       | 920                | 762       |
| Error Mean           | 12.03              | 8.58      | 13.33              | 11.04     |

experiment is to analyze the time and resources consumed by the pipeline when operating with different UAV numbers. The former mainly analyzes the time-consuming of three parts, while the latter mainly analyzes the VRAM requirements of three parts.

Experimental results show that as the number of drones increases, the total time consumption of the pipeline increases, as shown in Fig. 7 (a). When one, two and three UAVs operate simultaneously, it takes approximately 0.18 s, 0.33 s and 0.44 s for an image to be processed by the pipeline respectively. Fig. 7(b), (c) and (d) respectively show the time-consuming situation of each part of the system under multi-UAV. When three UAVs work simultaneously, the detection time increases from 0.06 s to 0.14 s, the enhancement time increases from 0.05 s to 0.09 s, and the tracking time increases from 0.03 s to 0.15 s. The results show that when multiple drones work simultaneously, the pipeline still maintains high real-time performance.

As shown in Fig. 8, the VRAM consumed by the pipeline increases as the number of UAVs increases. When operating with a single, two, and three UAVs, the VRAM consumed by the pipeline is approximately 4.95 g, 9.90 g, and 14.85 g respectively. When a UAV is added for operation, the increase in VRAM for the detection counting part and enhancement part is approximately 1.17 g and 3.78 g respectively. Experimental results show that if hardware resources are sufficient, the proposed pipeline can carry out multi-UAV operations for efficient operations.

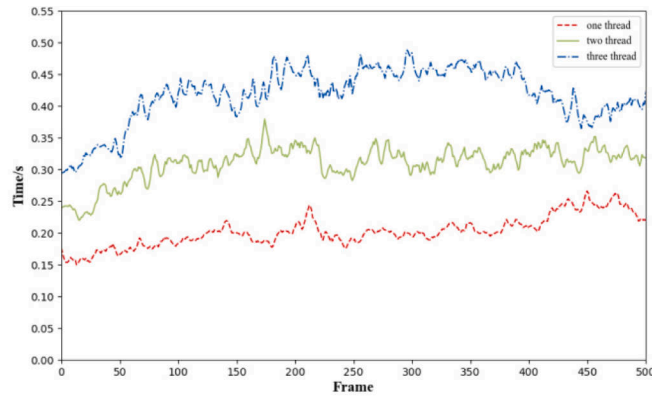
#### 4.4.4. Failure cases

During the counting process, an obvious phenomenon appeared that is very likely to be the cause of the increase in errors: During the transmission of UAV images, when the network condition is poor, image distortion or image data packet loss often occurs. The actual positions of some fruits are inconsistent with the positions predicted by the tracker. The fundamental reason is that WIFI is used to transmit data in this study, which has the disadvantage of network instability, resulting in data packet loss. The retransmission mechanism retransmits data packets when they are lost, thereby increasing latency. This is also the reason why the online counting method has a delay of 7 s ~ 10s. Using high-quality communication equipment can effectively improve the quality and stability of communication signals. Adaptive bitrate control technology can also be used to dynamically adjust the data transmission rate based on current network conditions to ensure a stable connection in unstable network environments.

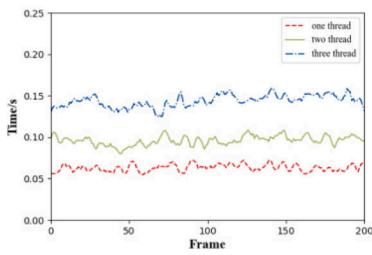
**Table 2**

Comparison of local performance.

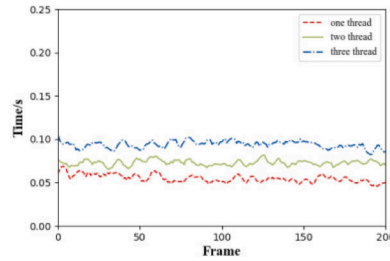
| Scenarios (kLux) | Videos | $FN_t$     |      | $FP_t$     |      | ID switch  |      | MOTA       |      | GT     |
|------------------|--------|------------|------|------------|------|------------|------|------------|------|--------|
|                  |        | YOLO_DSORT | Ours | YOLO_DSORT | Ours | YOLO_DSORT | Ours | YOLO_DSORT | Ours |        |
| Daytime/(>15)    | 1      | 57         | 31   | 185        | 115  | 134        | 118  | 0.76       | 0.83 | 1593   |
|                  | 2      | 27         | 23   | 131        | 64   | 71         | 47   | 0.75       | 0.85 | 910    |
|                  | 3      | 14         | 19   | 111        | 78   | 76         | 38   | 0.82       | 0.88 | 1112   |
|                  | 4      | 92         | 71   | 146        | 134  | 51         | 46   | 0.78       | 0.82 | 1377   |
|                  | 5      | 31         | 30   | 162        | 96   | 81         | 92   | 0.74       | 0.80 | 1073   |
| Evening/(<10)    | 6      | 18         | 7    | 129        | 127  | 91         | 68   | 0.85       | 0.87 | 1581   |
|                  | 7      | 16         | 10   | 134        | 117  | 56         | 42   | 0.78       | 0.82 | 939    |
|                  | 8      | 8          | 2    | 182        | 134  | 78         | 82   | 0.84       | 0.87 | 1689   |
|                  | 9      | 20         | 10   | 113        | 56   | 62         | 41   | 0.85       | 0.92 | 1261   |
|                  | 10     | 46         | 9    | 92         | 72   | 91         | 61   | 0.85       | 0.90 | 1487   |
| Mean             |        | 32.9       | 21.2 | 138.5      | 99.3 | 79.1       | 63.7 | 0.80       | 0.86 | 1302.2 |



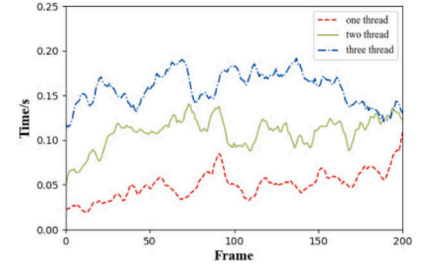
(a) Pipeline



(b) Detection



(c) Enhancement



(d) Tracking

Fig. 7. Time consumption of pipeline for multiple UAVs.

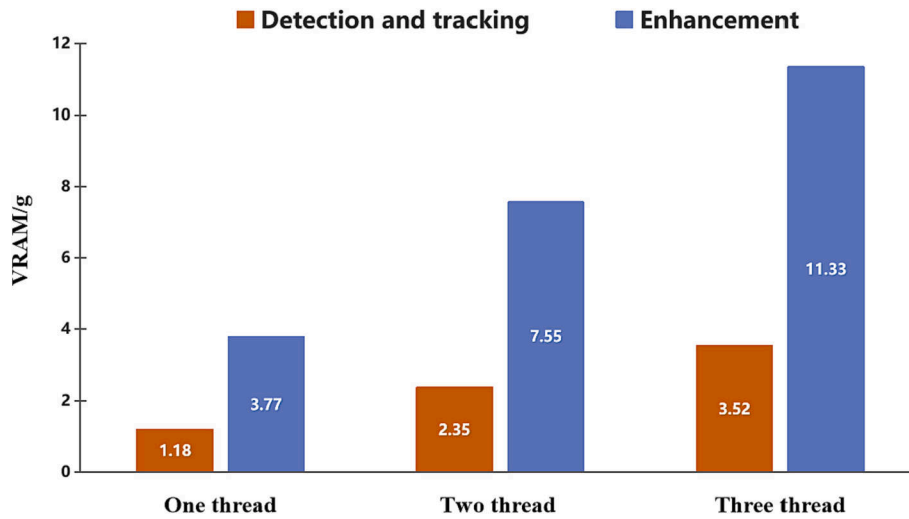


Fig. 8. The VRAM consumption of pipeline for multiple UAVs.

As shown in Table 2, the main two factors affecting counting performance are FP and ID switch. The increase in FP is mainly caused by the misidentification of leaves, while the increase in ID switch is caused by the failure of the pipeline to track fruits. Fig. 9(a) and (b) are the results of fruit ID before and after the appearance of stone pillars. ID switches mainly occur in two situations. In Fig. 9(c), ID switches do not result in overcounting. The reason is that fruits have similar appearance characteristics and are close to each other. When reappearing, the fruit may be matched to another counted target's trajectory. The overcounting of fruits is mainly caused by the ID switches in Fig. 9(d). In this study, fruits failed to be tracked due to the presence of obstacles

(such as passing people, stone pillars, etc.). In addition, although fruits that are severely obscured by branches and leaves can be observed from different angles while the UAV is moving, a small part of fruits will inevitably be missed.

## 5. Discussion

This paper proposes a novel solution for fruit counting about UAV applications in dark light scenes, especially the UAV live broadcast platform, low-light image enhancement, and detection and counting of fruits.

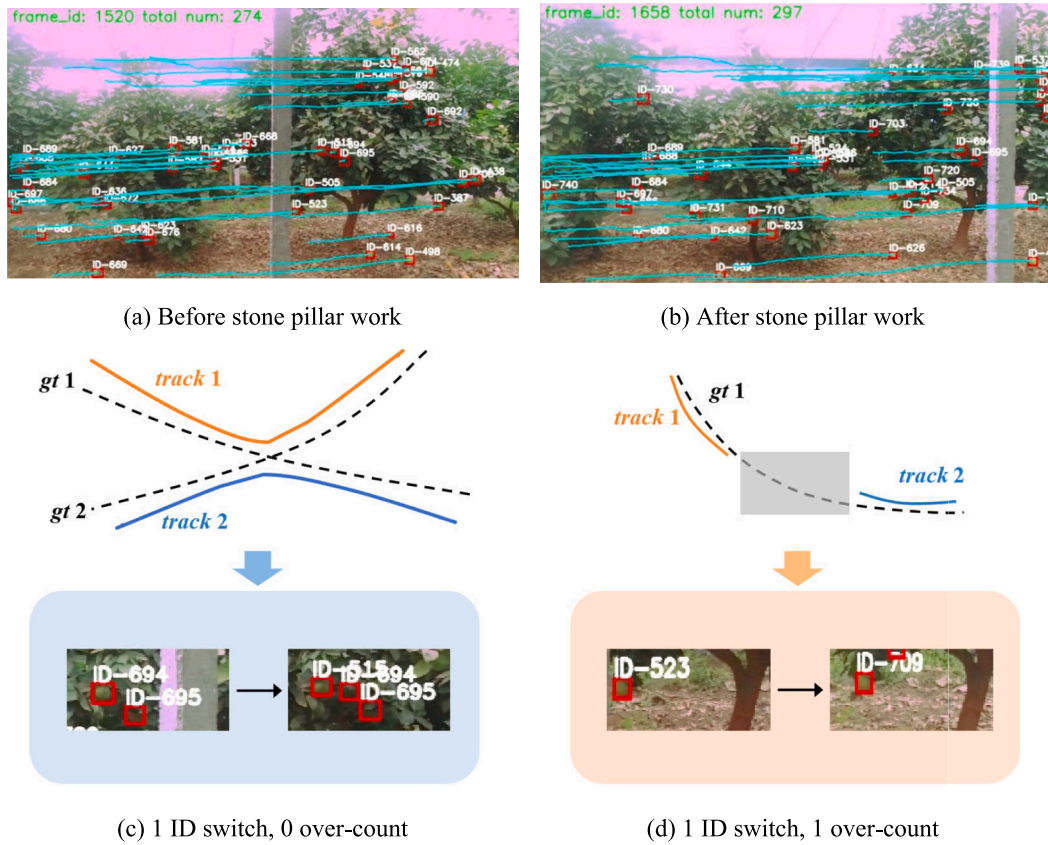


Fig. 9. The situations leading to ID switches and over-count.

Some previous studies have suggested that UAVs can be very effective in monitoring and counting tasks in the agricultural sector (Arakawa et al., 2023; Zhang et al., 2023; Apolo-Apolo et al., 2020a, 2020b; Méndez et al., 2019). Compared with the work of Arakawa et al. (2023) and Zhang et al. (2023), our study also built a UAV data transmission platform for orchards and developed an APP called FlyCounter to implement an end-to-end counting pipeline. Apolo-Apolo et al., 2020a, 2020b and Gené-Mola et al. (2020) obtained higher counting accuracy by reconstructing orchards. This study achieved low-cost fruit counting based on tracking-by-detection while ensuring accuracy. Our method does not require step-by-step procedures and is simple to operate, which can significantly reduce the processing time required to obtain accurate results. Pipelines facilitate the rapid development, iterative optimization and application of systems to help farmers develop optimal strategies as early as possible.

In the fruit detection task, compared with Zhang et al. (2022a, 2022b) and Chen et al. (2022), the improved Fruit-YOLO can effectively improve the detection performance. In particular, in response to the problem of poor robustness in low-illumination scenes in existing works such as Xu et al. (2023), Zhou et al. (2022) and Wang and He, 2022, this study innovatively introduces the IAT network to optimize detector, which further improves the accuracy metric of the system. Similar to the idea of using an image equalization algorithm (Zhou et al., 2022) and artificial light source equipment (Liang et al., 2020; Fu et al., 2021; Zhang et al., 2023), our system has better robustness and practical significance. In addition, the exploration of low illumination can help extend the system's operating time and can be extended to all-weather yield estimation and picking of other fruits in the orchard.

In the fruit counting task, compared with the existing state-of-the-art citrus counting work (Zhang et al., 2022a, 2022b), this paper uses more precise evaluation indicators (MOTA and MAPE) to evaluate the local performance and overall performance of the system respectively.

Evaluate. The proposed method outperforms the method proposed by Zhang et al., 2022a with lower ID Switch and MAPE. In addition, our system supports parallel operations of multiple UAVs, which has the advantage of efficient operations and promotion value.

Nevertheless, the proposed system still has the following limitations: (1) The UAV transmission platform cannot work properly when the network condition is poor, resulting in fruit tracking failure. Improving the communication quality and stability of the platform is a problem that needs to be addressed in the future. (2) After using the IAT network, the Fruit-YOLO network will miss or misdetect fruits. Further improving the IAT network can effectively address this problem, such as developing an image low-light area evaluation algorithm. (3) The pipeline only counts from one side of the fruit tree, which has problems with ID switching and missed detection. Research on cross-camera fruit counting based on multi-UAV collaborative operations can be considered in the future.

## 6. Conclusions

This paper proposes a new fruit-counting pipeline based on UAV, which can efficiently and robustly count fruits in large-scale orchards. To the best of our knowledge, this is the first work to realize an online and efficient counting pipeline for orchards in low-light scenarios based on UAVs. We can draw the following conclusions:

1. A UAV data transmission platform for orchards was built to obtain video streams online. Correspondingly, a flight control APP called FlyCounter was developed to achieve stable flight of UAVs.
2. The SPD-Conv is used in Fruit-YOLO's Backbone and Neck. Experimental results show that Fruit-YOLO is superior to YOLOv5 in terms of detection accuracy, and demonstrates remarkable performance

and effectively addresses issues related to missed detections and false detections.

- By introducing the IAT network, the problem of poor detection robustness under backlight during the day and low illumination conditions in the evening is effectively overcome.
- After obtaining the location information of fruits in the image, DeepSORT is used to track and count fruits. Experimental indicators show that the proposed method exhibits excellent robustness and superior counting performance in actual daytime and evening scenes in the orchard.

Field experiments were carried out to reflect the counting performance of pipeline under real operating scenarios. The time and resources consumed by the system under parallel operations with multiple UAVs were analyzed.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.agry.2024.103867>.

### CRedit authorship contribution statement

**Zhenhui Zheng:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Meng Wu:** Writing – original draft, Software. **Ling Chen:** Writing – original draft, Software. **Chenglin Wang:** Funding acquisition, Project administration, Resources, Writing – review & editing. **Juntao Xiong:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Lijiao Wei:** Writing – review & editing, Software. **Xiaoman Huang:** Visualization, Conceptualization. **Shuo Wang:** Visualization, Conceptualization. **Weihua Huang:** Visualization, Conceptualization. **Dongjie Du:** Visualization, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

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